Midterm Skills Exam	
Course Code: CPE212	Program: BSCPE
Course Title: Automating Server Management	Date Performed: 03/12/2026
Section: CPE31S1	Date Submitted: 03/12/2026
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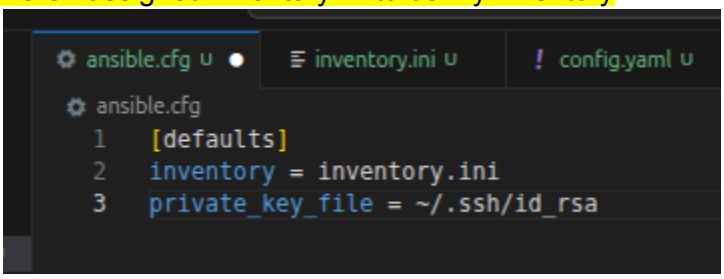
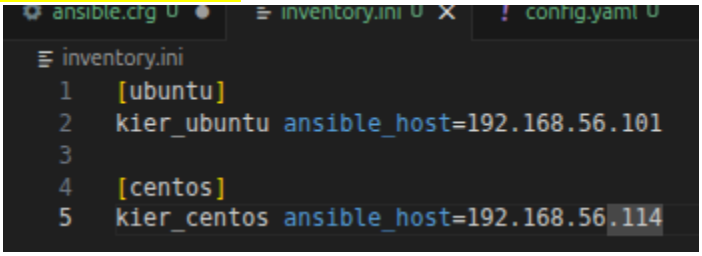
Output:

Here I assigned inventory.ini to be my inventory



And this is the contents of my inventory.ini. These are the ip addresses of my 2 other VMs, ubuntu and centos



For my config.yaml, this contains the necessary packages and commands that are needed for the other tasks to run

config.yaml > [] lamp_web_packages_ubuntu

```yaml
---
common_packages_ubuntu:
  - apt-transport-https
  - wget
  - curl
  - gnupg
  - software-properties-common

common_packages_centos:
  - wget
  - curl
  - python3

elastic_java_package_ubuntu: openjdk-11-jdk
elastic_java_package_centos: java-11-openjdk
elastic_repo_url: "deb https://artifacts.elastic.co/packages/8.x/apt stable main"
elastic_repo_key: https://artifacts.elastic.co/GPG-KEY-elasticsearch
elastic_repo_description_centos: Elastic repository for 8.x packages
elastic_repo_baseurl_centos: https://artifacts.elastic.co/packages/8.x/yum

nagios_packages_ubuntu:
  - nagios4
  - nagios-plugins-contrib
  - apache2

lamp_web_packages_ubuntu:
  - apache2
  - php
  - libapache2-mod-php
  - php-mysql

lamp_web_packages_centos:
  - httpd
  - php
  - php-mysqlnd

lamp_db_packages_ubuntu:
  - mariadb-server

lamp_db_packages_centos:
  - mariadb-server
```

Elastic_Stack

Here, I

```yaml
---
- name: Install Java for Elastic Stack on Ubuntu
  apt:
    name: "{{ elastic_java_package_ubuntu }}"
    state: present
    update_cache: yes
  when: ansible_facts['distribution'] == 'Ubuntu'

- name: Install Java for Elastic Stack on CentOS
  dnf:
    name: "{{ elastic_java_package_centos }}"
  when: ansible_distribution == 'CentOS'

- name: Add Elastic GPG key
  apt_key:
    url: "{{ elastic_repo_key }}"
    state: present
  when: ansible_facts['distribution'] == 'Ubuntu'

- name: Add Elastic repository
  apt_repository:
    repo: "{{ elastic_repo_url }}"
    state: present
    filename: elastic
  when: ansible_facts['distribution'] == 'Ubuntu'

- name: Add Elastic repository on CentOS
  yum_repository:
    name: elastic-8.x
    description: "{{ elastic_repo_description_centos }}"
    baseurl: "{{ elastic_repo_baseurl_centos }}"
    gpgcheck: yes
    gpgkey: "{{ elastic_repo_key }}"
    enabled: yes
  when: ansible_distribution == 'CentOS'

- name: Install requested Elastic component
  apt:
    name: "{{ elastic_component }}"
    state: present
    update_cache: yes
  when: ansible_facts['distribution'] == 'Ubuntu'

- name: Install requested Elastic component on CentOS
  dnf:
    name: "{{ elastic_component }}"
    state: present
    update_cache: yes
  when: ansible_distribution == 'CentOS'

- name: Ensure Elasticsearch logs directory exists on CentOS
  file:
    path: /usr/share/elasticsearch/logs
    state: directory
    owner: elasticsearch
    group: elasticsearch
    mode: '0755'
  when:
    - ansible_distribution == 'CentOS'
    - elastic_component == 'elasticsearch'

- name: Ensure Elasticsearch package directories exist
  file:
    path: "{{ item }}"
    state: directory
    owner: elasticsearch
    group: elasticsearch
    mode: '0755'
  loop:
    - /var/lib/elasticsearch
    - /var/log/elasticsearch
  when:
    - elastic_component == 'elasticsearch'

- name: Remove stale Elasticsearch secure settings from keystore
  shell: |
    for key in \
      autoconfiguration.password_hash \
      xpack.security.http.ssl.keystore.secure_password \
      xpack.security.transport.ssl.keystore.secure_password \
      xpack.security.transport.ssl.truststore.secure_password
    do
      /usr/share/elasticsearch/bin/elasticsearch-keystore list | grep -qx "$key" && \
      /usr/share/elasticsearch/bin/elasticsearch-keystore remove "$key" || true
    done
  when:
    - elastic_component == 'elasticsearch'
  notify: Restart Elasticsearch

- name: Configure Elasticsearch network settings
  lineinfile:
    path: /etc/elasticsearch/elasticsearch.yml
    regexp: '^{{ item.key }}:'
    line: "{{ item.key }}: {{ item.value }}"
  loop:
    - { key: 'network.host', value: '0.0.0.0' }
    - { key: 'discovery.type', value: 'single-node' }
    - { key: 'path.data', value: '/var/lib/elasticsearch' }
    - { key: 'path.logs', value: '/var/log/elasticsearch' }
    - { key: 'xpack.security.enabled', value: 'false' }
    - { key: 'xpack.security.http.ssl.enabled', value: 'false' }
    - { key: 'xpack.security.transport.ssl.enabled', value: 'false'
  when:
    - elastic_component == 'elasticsearch'
  notify: Restart Elasticsearch

- name: Configure Kibana server host
  lineinfile:
    path: /etc/kibana/kibana.yml
    regexp: '^server.host:'
    line: 'server.host: "0.0.0.0"'
  when:
    - elastic_component == 'kibana'
  notify: Restart Kibana

- name: Configure Kibana Elasticsearch host
  lineinfile:
    path: /etc/kibana/kibana.yml
    regexp: '^elasticsearch.hosts:'
    line: 'elasticsearch.hosts: ["http://127.0.0.1:9200"]'
  when:
    - elastic_component == 'kibana'
  notify: Restart Kibana

- name: Configure Logstash output
  copy:
    dest: /etc/logstash/conf.d/02-beats-input.conf
    content: |
      input {
        beats {
          port => 5044
        }
      }
      output {
        elasticsearch {
          hosts => ["http://127.0.0.1:9200"]
        }
      }
    owner: root
    group: root
    mode: '0644'
  when:
    - elastic_component == 'logstash'
  notify: Restart Logstash

- name: Enable and start requested Elastic service
  systemd:
    name: "{{ elastic_component }}"
    enabled: yes
    state: started
```

Lamp

```yaml
---
- name: Install Apache and PHP packages on Ubuntu web host
  apt:
    name: "{{ lamp_web_packages_ubuntu }}"
    state: present
    update_cache: yes
  when:
    - lamp_component == 'web'
    - ansible_facts['distribution'] == 'Ubuntu'

- name: Install Apache and PHP packages on CentOS web host
  dnf:
    name: "{{ lamp_web_packages_centos }}"
    state: present
    update_cache: yes
  when:
    - lamp_component == 'web'
    - ansible_distribution == 'CentOS'

- name: Create PHP test page
  copy:
    dest: /var/www/html/index.php
    content: |
      <?php
      phpinfo();
      ?>
    owner: root
    group: root
    mode: '0644'
  when:
    - lamp_component == 'web'
    - ansible_facts['distribution'] in ['Ubuntu', 'CentOS']

- name: Enable and start Apache web service on Ubuntu
  systemd:
    name: apache2
    enabled: yes
    state: started
  when:
    - lamp_component == 'web'
    - ansible_facts['distribution'] == 'Ubuntu'

- name: Enable and start Apache web service on CentOS
  systemd:
    name: httpd
    enabled: yes
    state: started
  when:
    - lamp_component == 'web'
    - ansible_distribution == 'CentOS'

- name: Install MariaDB on Ubuntu database host
  apt:
    name: "{{ lamp_db_packages_ubuntu }}"
    state: present
    update_cache: yes
  when:
    - lamp_component == 'db'
    - ansible_facts['distribution'] == 'Ubuntu'

- name: Enable and start MariaDB on Ubuntu database host
  systemd:
    name: mariadb
    enabled: yes
    state: started
  when:
    - lamp_component == 'db'
    - ansible_facts['distribution'] == 'Ubuntu'

- name: Install MariaDB on CentOS database host
  dnf:
    name: "{{ lamp_db_packages_centos }}"
    state: present
    update_cache: yes
  when:
    - lamp_component == 'db'
    - ansible_distribution == "CentOS"
- name: Enable and start MariaDB on CentOS database host
  systemd:
    name: mariadb
    enabled: yes
    state: started
  when:
    - lamp_component == 'db'
    - ansible_distribution == "CentOS"
```

Nagios

```yaml
---
- name: Install Nagios packages on Ubuntu host
  apt:
    name: "{{ nagios_packages_ubuntu }}"
    state: present
    update_cache: yes
  when: ansible_facts['distribution'] == 'Ubuntu'

- name: Enable and start Nagios on Ubuntu
  systemd:
    name: nagios4
    enabled: yes
    state: started
  when: ansible_facts['distribution'] == 'Ubuntu'

- name: Enable and start Apache for Nagios UI on Ubuntu
  systemd:
    name: apache2
    enabled: yes
    state: started
  when: ansible_facts['distribution'] == 'Ubuntu'
```

Playbook.yml

Ansible Playbook - Ansible playbook files (ansible.json)

```yaml
---
- name: Install Elasticsearch on Ubuntu and CentOS hosts
  hosts: all
  become: true
  vars_files:
    - config.yaml
  vars:
    elastic_component: elasticsearch
  roles:
    - Elastic_stack

- name: Install Kibana on Ubuntu and CentOS hosts
  hosts: all
  become: true
  vars_files:
    - config.yaml
  vars:
    elastic_component: kibana
  roles:
    - Elastic_stack

- name: Install Logstash on Ubuntu and CentOS hosts
  hosts: all
  become: true
  vars_files:
    - config.yaml
  vars:
    elastic_component: logstash
  roles:
    - Elastic_stack

- name: Install Nagios on Ubuntu
  hosts: ubuntu
  become: true
  vars_files:
    - config.yaml
  roles:
    - nagios

- name: Install Apache and PHP on Ubuntu and CentOS hosts
  hosts: all
  become: true
  vars_files:
    - config.yaml
  vars:
    lamp_component: web
  roles:
    - lamp

- name: Install MariaDB on Ubuntu and CentOS hosts
  hosts: all
  become: true
  vars_files:
    - config.yaml
  vars:
    lamp_component: db
  roles:
    - lamp
```

Verifying:
Ubuntu:

CentOS:

GitHub commands:

```
kier@LocalMachine:~/CPE_MIDEXAM_BONITA$ git add .
kier@LocalMachine:~/CPE_MIDEXAM_BONITA$ git commit -m "Midterm Exam"
[main 37e4a0c] Midterm Exam
 28 files changed, 710 insertions(+)
 create mode 100644 Elastic_stack/README.md
 create mode 100644 Elastic_stack/defaults/main.yml
 create mode 100644 Elastic_stack/handlers/main.yml
 create mode 100644 Elastic_stack/meta/main.yml
 create mode 100644 Elastic_stack/tasks/main.yml
 create mode 100644 Elastic_stack/tests/inventory
 create mode 100644 Elastic_stack/tests/test.yml
 create mode 100644 Elastic_stack/vars/main.yml
 create mode 100644 ansible.cfg
 create mode 100644 config.yaml
 create mode 100644 inventory.ini
 create mode 100644 lamp/README.md
 create mode 100644 lamp/defaults/main.yml
 create mode 100644 lamp/handlers/main.yml
 create mode 100644 lamp/meta/main.yml
 create mode 100644 lamp/tasks/main.yml
 create mode 100644 lamp/tests/inventory
 create mode 100644 lamp/tests/test.yml
 create mode 100644 lamp/vars/main.yml
 create mode 100644 nagios/README.md
 create mode 100644 nagios/defaults/main.yml
 create mode 100644 nagios/handlers/main.yml
 create mode 100644 nagios/meta/main.yml
 create mode 100644 nagios/tasks/main.yml
 create mode 100644 nagios/tests/inventory
 create mode 100644 nagios/tests/test.yml
 create mode 100644 nagios/vars/main.yml
 create mode 100644 playbook.yml
kier@LocalMachine:~/CPE_MIDEXAM_BONITA$ git push
Enumerating objects: 45, done.
Counting objects: 100% (45/45), done.
Delta compression using up to 4 threads
Compressing objects: 100% (22/22), done.
Writing objects: 100% (44/44), 6.37 KiB | 2.12 MiB/s, done.
Total 44 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), done.
To github.com:knmbonita/CPE_MIDEXAM_BONITA.git
   27e9249..37e4a0c  main -> main
kier@LocalMachine:~/CPE_MIDEXAM_BONITA$
```

GitHub Link:

https://github.com/knmbonita/CPE_MIDEXAM_BONITA#

Conclusion:

In this activity, I successfully created an ansible playbook for the task that was given to us but unfortunately, I wasn't able to finish the process to run the playbook tasks because of a computer problem. Overall, I'm confident that it will run if the computer

was working properly. I will try to lower down the specs next activity.